

NETWORK PROGRAMMING LABORATORY

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webanatomy: Anatomy of a Web Request

Write a Rust program that observes what happens at the TCP level while a single web request is performed.

The program receives a URL, a network interface, and a timeout. It performs one HTTP GET request using `ureq` and, at the same time, captures TCP packets with `pcap`. At the end, it prints a compact report that compares application-level information with packet-level counters.

Command Line

The program must be invoked as follows:

```
sudo webanatomy <url> -i <interface> -t <timeout>
```

Examples:

```
sudo webanatomy http://example.com/ -i eth0 -t 5  
sudo webanatomy https://example.com/ -i en0 -t 5
```

The timeout is expressed in seconds. If omitted, a default value of 5 seconds may be used.

Required Behavior

The program must:

1. parse the command-line arguments;
2. parse the input URL using the `url` crate;
3. accept only `http` and `https` URLs;
4. extract the host and the remote port;
5. use port 80 for `http` and port 443 for `https`, unless the URL explicitly specifies another port;
6. resolve the host to one or more IPv4 addresses;
7. build a BPF filter that selects TCP packets to or from those IPv4 addresses and the selected remote port;
8. **start a packet capture on the selected interface;**
9. perform one HTTP GET request using `ureq`;

10. count TCP packets and bytes in the outgoing and incoming directions;
11. stop the capture when the timeout expires, or earlier if a TCP FIN or RST flag is observed;
12. print a final report.

Packet Capture

After resolving the URL host, install a BPF filter similar to:

```
(host 93.184.216.34) and tcp and port 443
```

If more than one IPv4 address is resolved, the filter may include all of them:

```
(host 93.184.216.34 or host 93.184.216.35) and tcp and port 443
```

For each captured Ethernet frame, the program should:

1. parse the Ethernet header and keep only IPv4 packets;
2. parse the IPv4 header;
3. parse the TCP header;
4. decide whether the packet is outgoing or incoming;
5. update the counters.

A packet is outgoing if its destination IPv4 address is one of the resolved addresses and its destination TCP port is the URL port.

A packet is incoming if its source IPv4 address is one of the resolved addresses and its source TCP port is the URL port.

Report

The final output should look like:

```
webanatomy
URL: https://example.com/
Host: example.com
Remote port: 443
Resolved IPv4 addresses: [93.184.216.34]

Application
HTTP status: 200
Bytes received by GET: 1256
Request time: 74.21 ms

Captured TCP
Outgoing packets: 8
Incoming packets: 10
Outgoing bytes: 2140
Incoming bytes: 8460
FIN observed: yes
RST observed: no
```